Seulki Han, Ph.D.

Postdoctoral Researcher, Department of Chemical & Biomolecular Engineering, University of Delaware

Email: seulkih@udel.edu — Google Scholar

Education & Training

University of Delaware (UDel) — Newark, DE, USA
Postdoctoral Researcher, Chemical and Biomolecular Engineering — Present
(Advisor: Dr. Marianthi Ierapetritou)

University of Connecticut (UConn) — Storrs, CT, USA
Ph.D., Chemical and Biomolecular Engineering (Advisor: Dr. George M. Bollas) — May 2025

Incheon National University (INU) — Incheon, South Korea
M.S., Energy and Chemical Engineering (Advisor: Dr. Jiyong Kim) — Feb 2017
B.S., Energy and Chemical Engineering — Feb 2015

Research Interests

- **Physics-Informed and Data-Driven Modeling**
  - Reduced-order, surrogate, and interpretable models for evolving engineering systems
- **Process Design, Operation, and Sustainability Assessment**
  - Process simulation, scale-up, optimization, techno-economic analysis (TEA), and life cycle assessment (LCA)
- **Energy and Resource Systems Optimization**
  - Supply chain design, infrastructure planning, and multi-period optimization for low-carbon fuels and circular resource systems

Research Experience

Postdoctoral Researcher — Jun 2025 – Present
ProOptiMA Lab, UDel

- Lead independent and collaborative research across multiple sponsored projects
- Translate experimental data into process and system models to support engineering decisions

- ***Project 1. FEC: Tri-State Collaborative for Defluorination of Per- and Polyfluoroalkyl Substances***
  Supported by NSF Focused EPSCoR Collaborations
  - Simulate alternative PFAS treatment trains integrating separation, concentration, destruction, and disposal
  - Compare treatment configurations through TEA and LCA to evaluate cost, energy use, and environmental impacts

- ***Project 2. Advanced Manufacturing of Renewable and Recyclable Polymers***
  Supported by NSF EPSCoR
  - Develop an integrated PA6 recycling process from experimental recovery data to evaluate scale-up feasibility
  - Identify process bottlenecks and benchmark solvent-based recycling against chemical recycling routes using TEA and LCA

- ***Project 3. Supply Chain Decision Making Framework Considering Uncertainty***
  Supported by NSF Award CBET-2217472
  - Develop a national-scale SAF supply chain model integrating seasonal feedstock availability, storage, existing infrastructure, and airport demand
  - Evaluate how resource seasonality and infrastructure shape facility utilization, regional sourcing, and SAF deployment

Graduate Research Assistant Aug 2019 – May 2025
UTC-IASE, UConn

- Conducted collaborative research on predictive monitoring for advanced manufacturing
- Developed models for state estimation and prognostics under changing operating conditions

- ***Project 1. Smart and Agile Aerospace Manufacturing***
  Supported by AFRL
  - Developed a physics-informed recursive symbolic regression framework for tool wear and remaining useful life prediction across various machining cases
  - Developed GAN-based data augmentation to improve tool wear classification using limited and imbalanced data

- ***Project 2. Energy Management Systems for Subtractive and Additive Precision Manufacturing***
  Supported by DOE–CESMII
  - Developed machine learning models to relate multimodal sensor signals to machine state and tool wear
  - Identified informative sensor combinations and features, demonstrating that selected sensors can outperform the full sensor set

Researcher Mar 2017 – Jun 2019
Innovation Center for Chemical Engineering (ICCE)

- Conducted research on sustainable energy, resource integration, and carbon utilization
- Contributed to a successful KRICT/MSIT research proposal on residue gas utilization
- Co-invented a patented optimization-based carbon utilization strategy

- ***Project 1. Optimization-Based Decision-Making Model for Residue Gas Utilization***
  Supported by Korea Research Institute of Chemical Technology (KRICT), Ministry of Science and ICT (MSIT)
  - Developed a process simulation and superstructure optimization framework for coke oven gas utilization
  - Identified how production objectives affect separation and conversion pathways for hydrogen, methane, methanol, and Fischer–Tropsch fuels

Graduate Research Assistant Mar 2015 – Feb 2017
PSE Lab, INU

- Conducted research on the design, operation, and investment planning of renewable energy supply systems

- ***Project 1. Development of a Decision Support Model for Design of Multi-Scaling Based Energy Systems***
  Supported by National Research Foundation of Korea (NRF), Ministry of Science and ICT (MSIT)
  - Developed multi-period MILP models for technology deployment, energy flows, and infrastructure investment
  - Evaluated how regional resource availability and interregional transport shape renewable energy supply strategies

Undergraduate Research Assistant Jan 2014 – Feb 2015

PSE Lab, INU

- Conducted foundational research on renewable energy system modeling and analysis

Journal Publications

In Preparation

- **Han, S.**, Yuliu, Z., and Ierapetritou, M. Life Cycle Assessment of Reusable Grocery Bags Based on Mechanical Durability and Reuse Cycles.
- **Han, S.**, Fields, C., Lobo, R., and Ierapetritou, M. Comparative Techno-Economic and Life Cycle Assessment of Solvent-based Nylon Recycling Systems.

Under Review

- **Han, S.** and Ierapetritou, M. A surrogate-driven optimization framework for volatile fatty acid (VFA) production and recovery from food waste. *Bioresource Technology*.
- **Han, S.** and Ierapetritou, M. A national-scale optimization framework for sustainable aviation fuel deployment. *Computers & Chemical Engineering*.
- **Han, S.** and Bollas, G. M. Enhancing multi-class tool wear classification via GAN-based scalogram augmentation. *Journal of Intelligent Manufacturing*.

Published Articles

10. Shapiro, A. J., O'Halloran, R. C., Tavares, D., Huynh, D., **Han, S.**, Shi, R., Gupta, Y., O'Dea, R., Haney, S., Otsuka, Y., Sadula, S., Vlachos, D. G., Ierapetritou, M., Levia, D. F., and Epps, T. H. (2026). Forest residue harvest optimization: Spanning the bridge between plant biology and biorefinery performance. *Green Chemistry*, 28(30), 12662–12676.
9. **Han, S.**, Awasthi, U., and Bollas, G. M. (2025). Physics-informed symbolic regression for tool wear and remaining useful life predictions in manufacturing. *Journal of Manufacturing Systems*, 80, 734–748.
8. **Han, S.**, Yang, Q., Pattipati, K. R., and Bollas, G. M. (2022). Sensor selection and tool wear prediction with data-driven models for precision machining. *Journal of Advanced Manufacturing and Processing*, 4(4).
7. **Han, S.**, Mannan, N., Stein, D. C., Pattipati, K. R., and Bollas, G. M. (2021). Classification and regression models of audio and vibration signals for machine state monitoring in precision machining systems. *Journal of Manufacturing Systems*, 61, 45–53.
6. You, C.†, **Han, S.**†, and Kim, J. (2021). Integrative design of the optimal biorefinery and bioethanol supply chain under the water-energy-food-land (WEFL) nexus framework. *Energy*, 228, 120574. (†*Equal contribution*)
5. **Han, S.** and Kim, J. (2019). A multi-period MILP model for the investment and design planning of a national-level complex renewable energy supply system. *Renewable Energy*, 141, 736–750.
4. **Han, S.**, Kim, S., Kim, Y. T., Kwak, G., and Kim, J. (2019). Optimization-based assessment framework for carbon utilization strategies: Energy production from coke oven gas. *Energy Conversion and Management*, 187, 1–14.
3. **Han, S.** and Kim, J. (2017). Optimization-based integration and analysis of a complex renewable energy system for the transportation sector. *Chemical Engineering Research and Design*, 128, 1–14.
2. **Han, S.** and Kim, J. (2017). An optimization model to design and analysis of renewable energy supply strategies for residential sector. *Renewable Energy*, 112, 222–234.
1. **Han, S.**, Won, W., and Kim, J. (2017). Scenario-based approach for design and comparative analysis of conventional and renewable energy systems. *Energy*, 129, 86–100.

Book Chapters

1. Cho, S., Won, W., **Han, S.**, Kim, S., You, C., and Kim, J. (2018). An optimization-based design and analysis of a biomass derived hydrogen energy system. *Computer Aided Chemical Engineering*, 44, 1573–1578.

Patents

1. **Han, S.**, Kim, S., and Kim, J. (2021). Republic of Korea. Patent No. 10-2203502. Method and server for establishing optimization-based carbon utilization strategy.

Conference Proceedings & Presentations

13. **Han, S.** and Bollas, G. M. (2025). cGAN-based feature representation and data augmentation for tool wear monitoring. *11th International Conference on Control, Decision and Information Technologies (CoDIT 2025)*, Split, Croatia.
12. **Han, S.**, Awasthi, U., and Bollas, G. M. (2025). Recursive modeling for generic tool wear prognostics in precision machining. *11th IFAC Conference on Manufacturing Modeling, Management and Control (IFAC MIM 2025)*, Trondheim, Norway.
11. **Han, S.**, Awasthi, U., and Bollas, G. M. (2024). Symbolic regression-based recursive surrogate model for a manufacturing process. *AIChE Annual Meeting*, San Diego, CA, USA.
10. **Han, S.**, Mishra, D., Pattipati, K. R., and Bollas, G. M. (2024). Symbolic regression-based hybrid models for a manufacturing process. *10th International Conference on Control, Decision and Information Technologies (CoDIT 2024)*, Valletta, Malta, 211–216.
9. Mishra, D., **Han, S.**, Pattipati, K. R., and Bollas, G. M. (2023). Explainable symbolic regression model for tool wear diagnosis. *9th International Conference on Control, Decision and Information Technologies (CoDIT 2023)*, Rome, Italy, 2139–2144.
8. **Han, S.** and Bollas, G. M. (2021). Regression model for tool wear monitoring in precision machining. *AIChE Annual Meeting*, Boston, MA, USA.
7. **Han, S.**, Maloney, T., and Bollas, G. M. (2020). Machine learning approach for tool condition monitoring in manufacturing systems. *AIChE Annual Meeting*, Virtual, USA.
6. **Han, S.** and Kim, J. (2019). An optimization model for design and analysis of land-water-energy-food nexus in a bioethanol supply chain. *KIChE Annual Meeting*, Jeju, Korea.
5. **Han, S.**, Kim, S., Kim, Y. T., Kwak, G., and Kim, J. (2018). MILP model for assessment of carbon utilization strategies. *KIChE Annual Meeting*, Daegu, Korea.
4. **Han, S.** and Kim, J. (2016). MILP model for technology integration and investment planning of a complex renewable energy system. *KIChE Annual Meeting*, Busan, Korea.
3. **Han, S.** and Kim, J. (2015). Optimization-based analysis of a complex energy superstructure. *ICCPE'15*, Barcelona, Spain.
2. **Han, S.** and Kim, J. (2015). Optimization-based analysis of a complex energy system using renewable energy sources. *KIChE Annual Meeting*, Jeju, Korea.
1. **Han, S.** and Kim, J. (2014). Design and life-cycle cost analysis of scenario-based energy supply system for transportation sector. *KIChE Annual Meeting*, Daejeon, Korea.

Teaching & Mentoring Experience

Teaching Assistant Aug 2019 – May 2020

University of Connecticut

- Advanced Process Control (Spring 2020)

- Fluid Mechanics (Fall 2019)

Teaching Assistant Aug 2015 – Dec 2016
Incheon National University

- Chemical Reaction Engineering (Fall 2015, 2016)

Research Mentoring Jun 2017 – May 2019
Innovation Center for Chemical Engineering

- Mentored 1 undergraduate and 2 master's students in modeling, data analysis, and manuscript preparation

Research Mentoring Jun 2015 – Dec 2016
Incheon National University

- Mentored 2 undergraduate students in optimization modeling, literature review, and research analysis

Honors & Awards

Pratt & Whitney Advanced Systems Engineering Fellowship 2024
Pratt & Whitney Advanced Systems Engineering Travel Fellowship 2024
Predoctoral Travel Award, University of Connecticut 2024
Best Oral Presentation Award, KIChE Conference 2019
Graduate Academic Scholarship, Incheon National University 2015–2016
Undergraduate Academic Scholarship, Incheon National University 2012–2014

Service

Reviewer, *Industrial & Engineering Chemistry Research* 2025–Present
Reviewer, *International Conference on Control, Decision and Information Technologies (CoDIT)* 2024
Reviewer, *Computers & Chemical Engineering* 2023–Present
Treasurer, UConn Korean Graduate Student Association 2020–2021